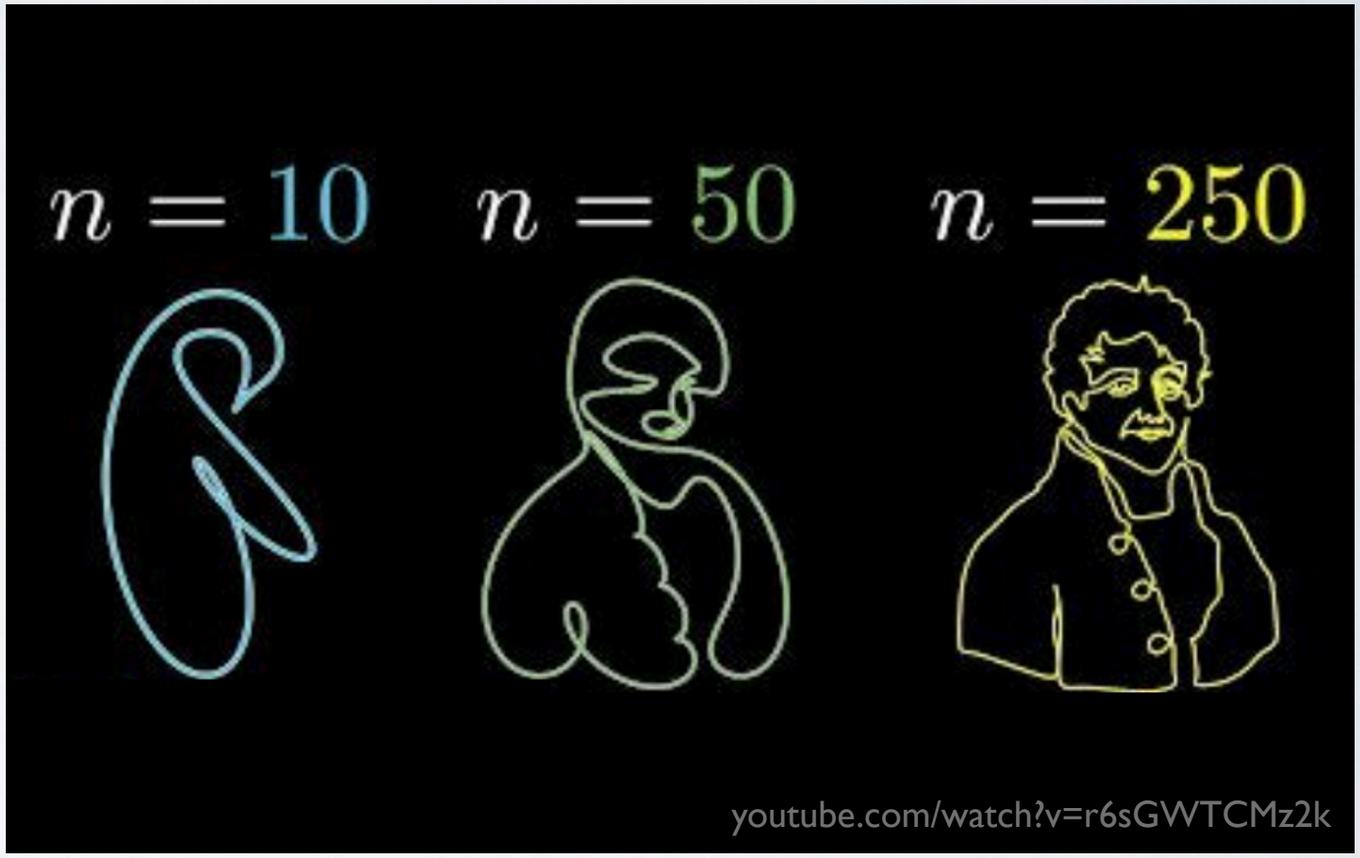
Week 9 - #1

Fourier Series and Transforms (I)





Today: Ch 7

Next Class: Ch 7

Ji-hoon Kim (Seoul National University)

This material is for the educational purpose only, and is not for public consumption or mass distribution without the author's explicit consent.
Despite his best efforts, the author apologizes for any inaccurate or outdated information and references.

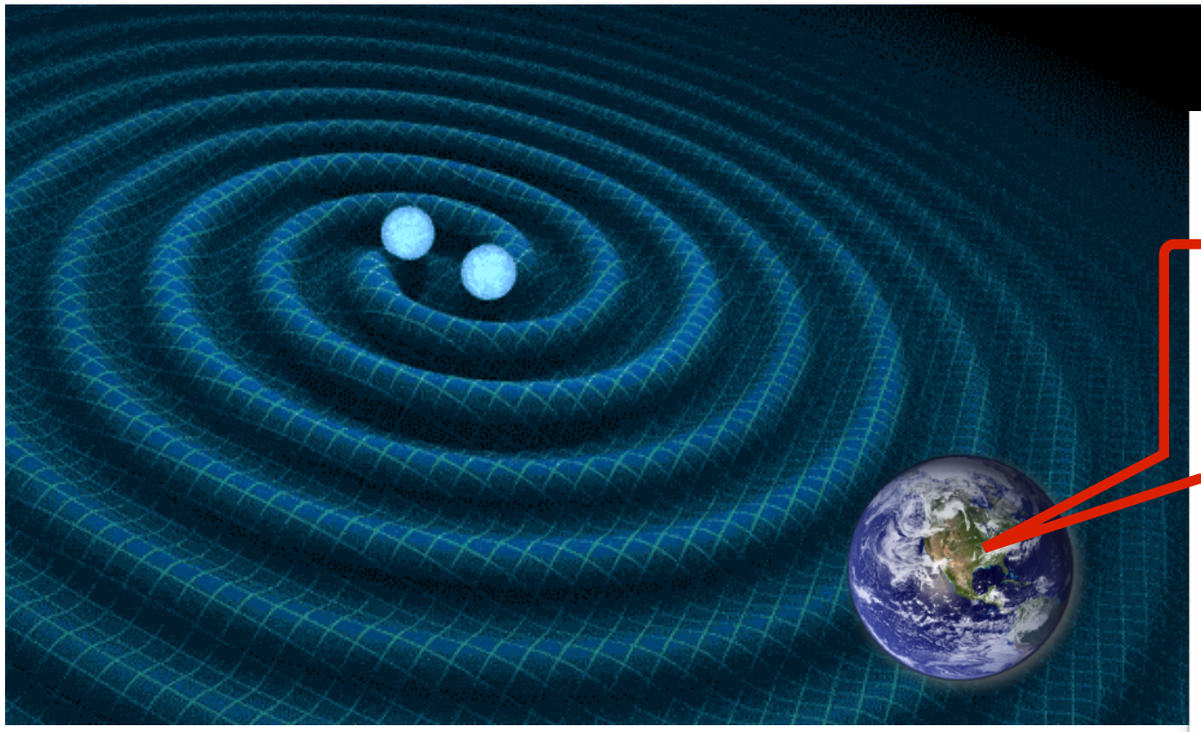
Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #12

— [open book and open note, but no cellphone or laptop, drop it off as you leave the class] —

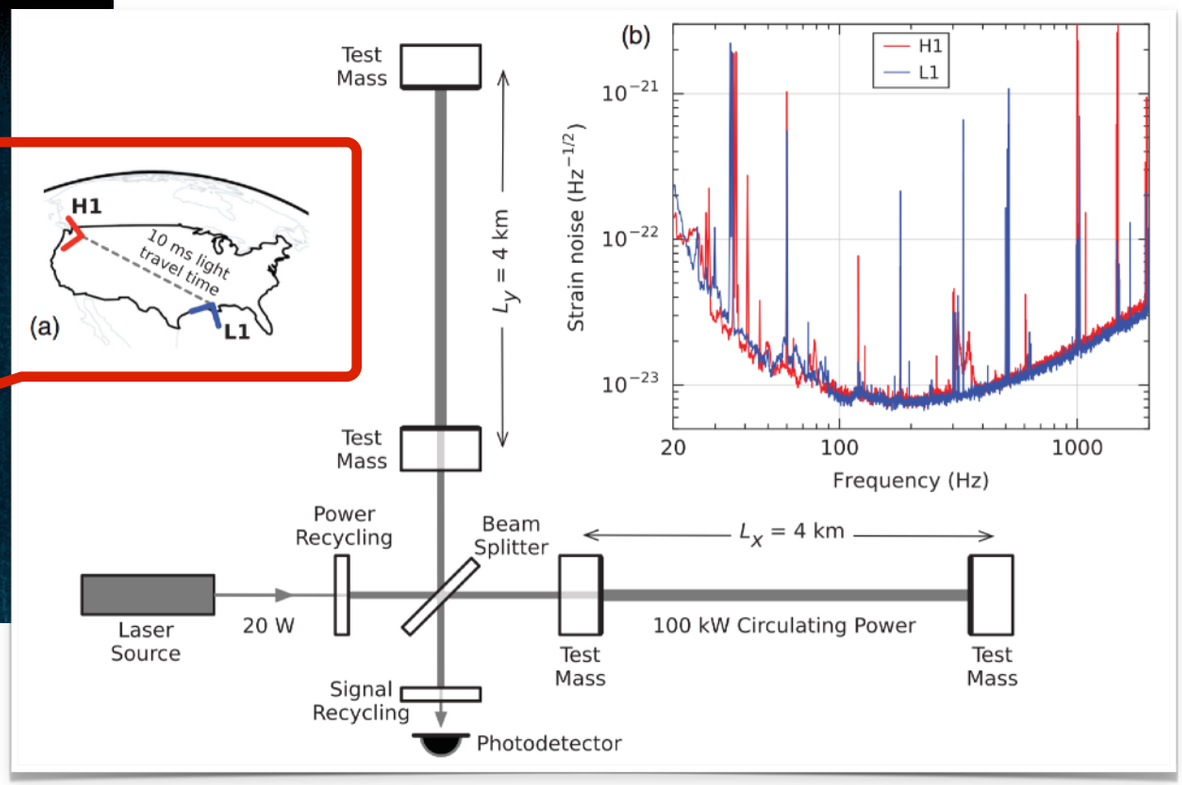
Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Boas, Chapter 7, Section 3, Problem 5 (see p.345/373 for definitions of *fundamentals* & *overtones*)

2. Boas, Chapter 7, Section 4, Problem 2 (which allows us to trivially prove Eq.(4.8) in p.349)



LIGO



Abbott et al. (2016a)

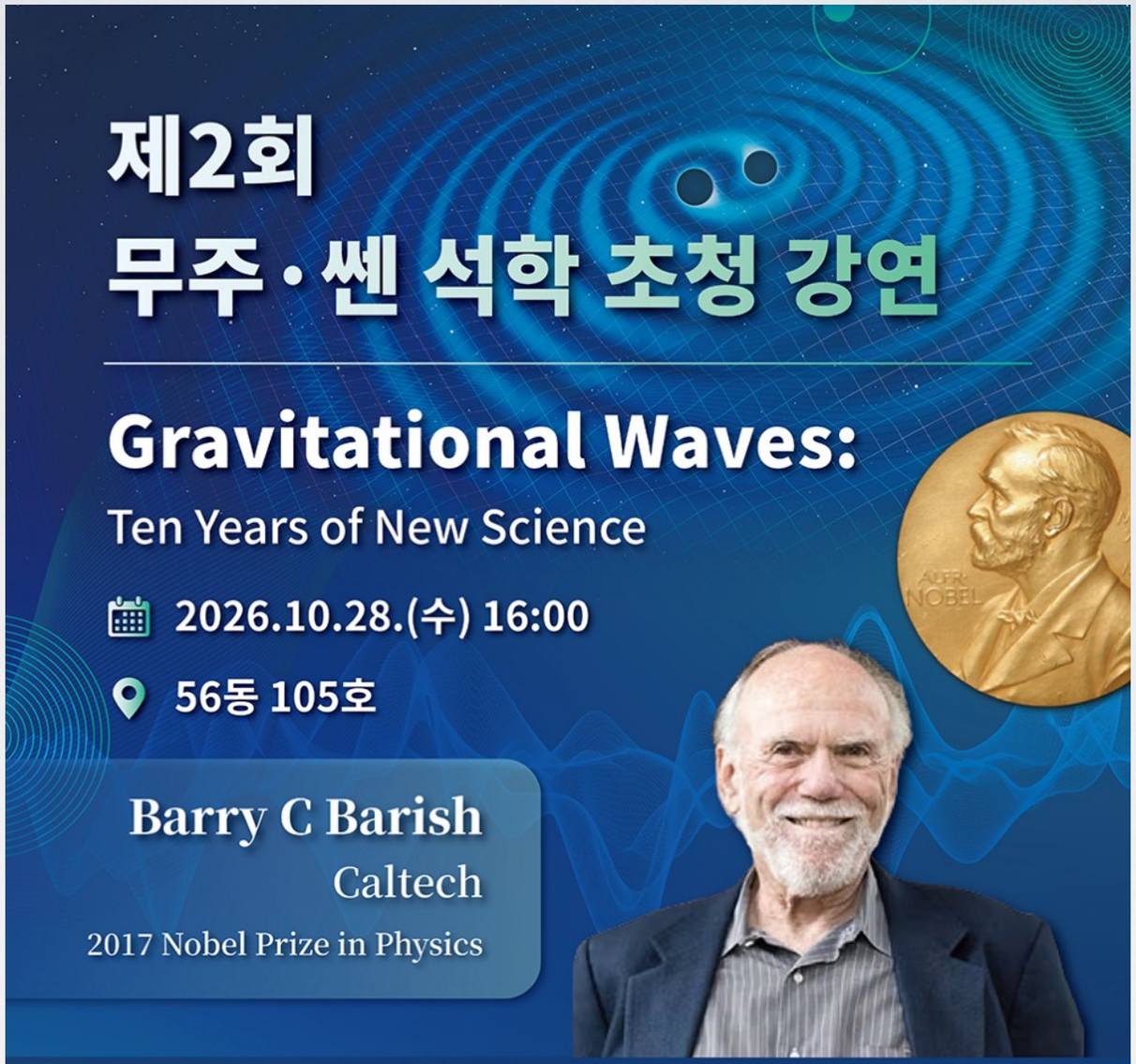
Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 7, Boas, 3rd ed.

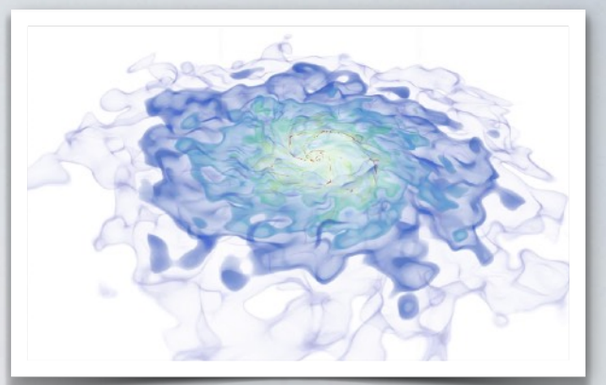
The problems I suggest you to take a deeper look into include, but are not limited to, the following. The class homework assignments will mainly be from this list.

- Section 02: Problems 12, 17
- Section 03: Problems 8
- Section 04: Problems 1, 13, 16
- Section 05: Problems 1, 3, 4, 7, 9, 11
- Section 06: Problems 10, 12 (for Problem 5.7 and 5.9), 14, 15
- Section 07: Problems 7, 9, 11, 13
- Section 08: Problems 2, 14, 15(a)(b), 20
- Section 09: Problems 9, 10, 11, 19, 20, 23
- Section 10: Problems 2
- Section 11: Problems 6, 8
- Section 12: Problems 6, 9, 10, 15, 18, 21, 23, 26, 31, 35
- Section 13: Problems 4, 8, 14, 23

48 problems / 218 total

Lecture By Nobel Laureates (무주 쎈 석학초청강연)

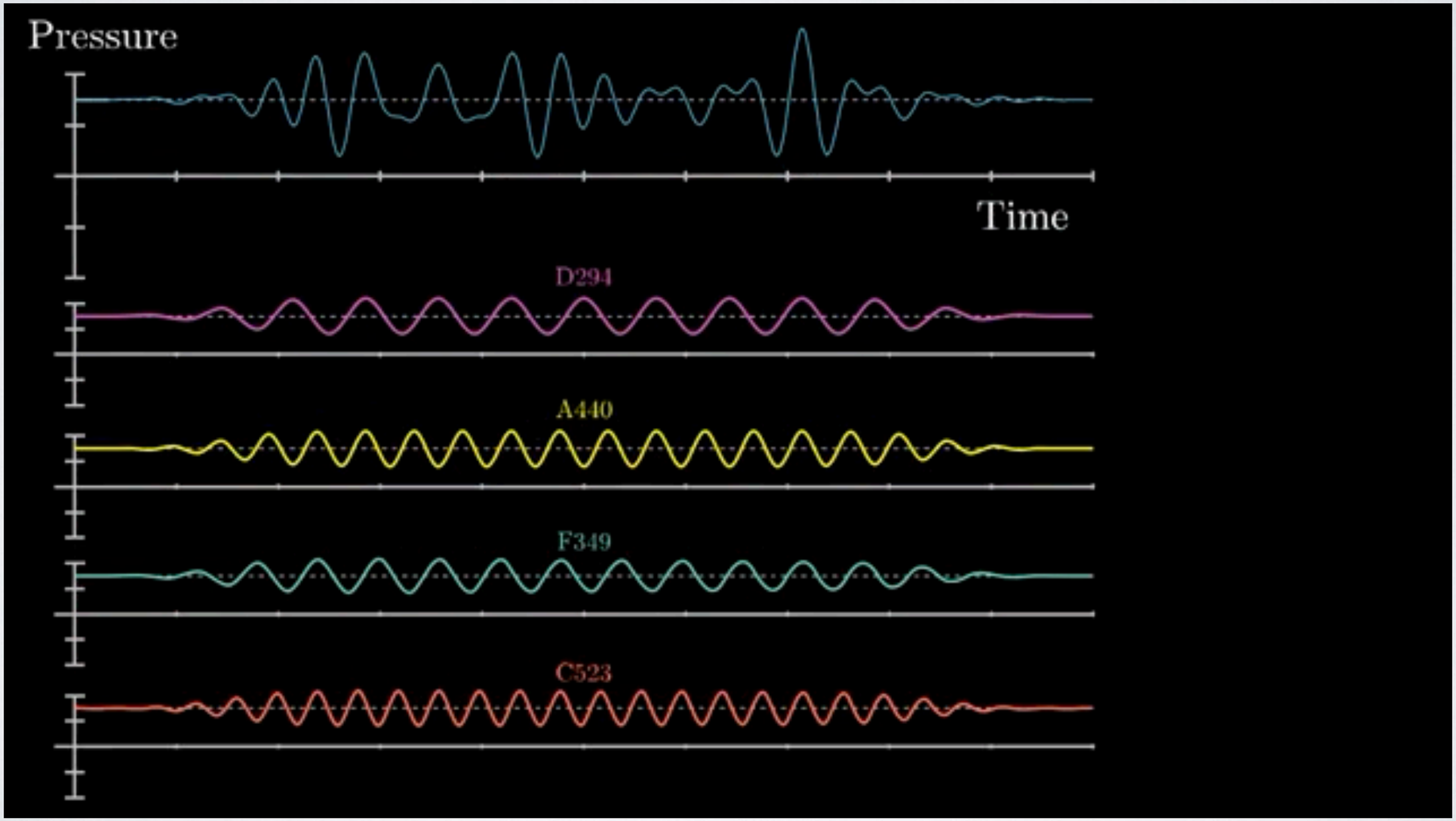




Fourier Series

Fourier Series

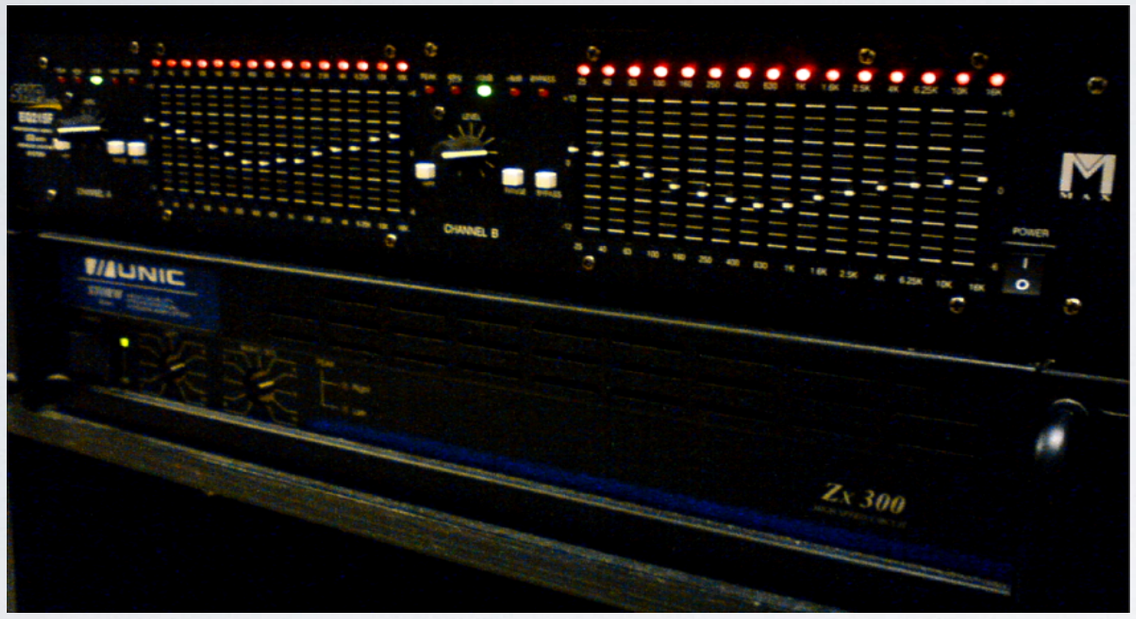
- A visualization of Fourier series.



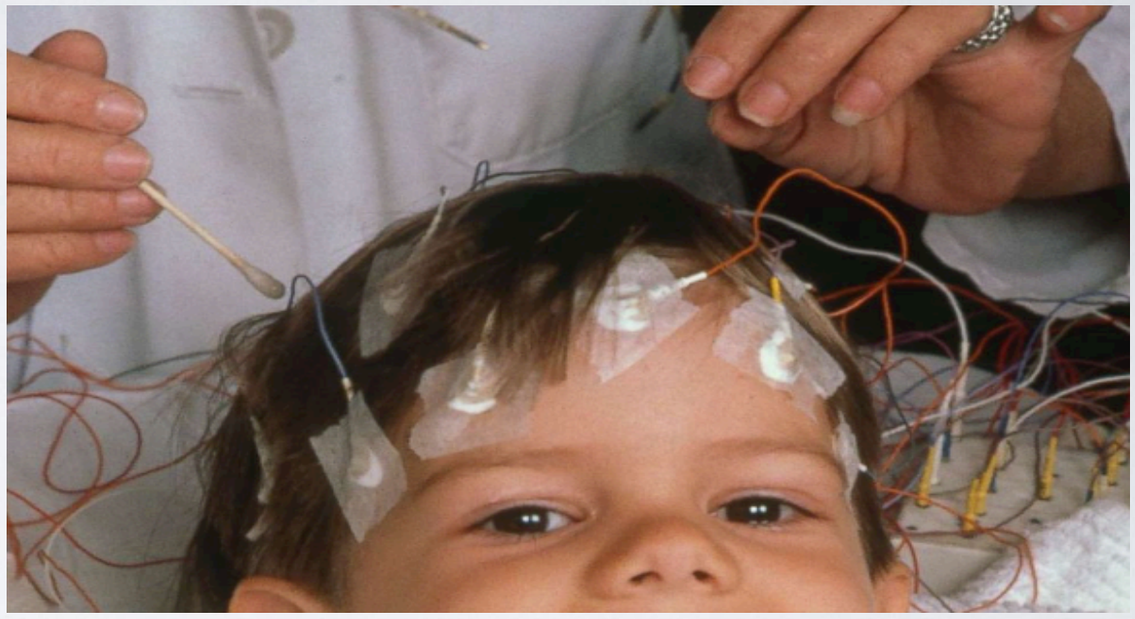
youtube.com/watch?v=spUNpyF58BY

Applications of Fourier Analysis

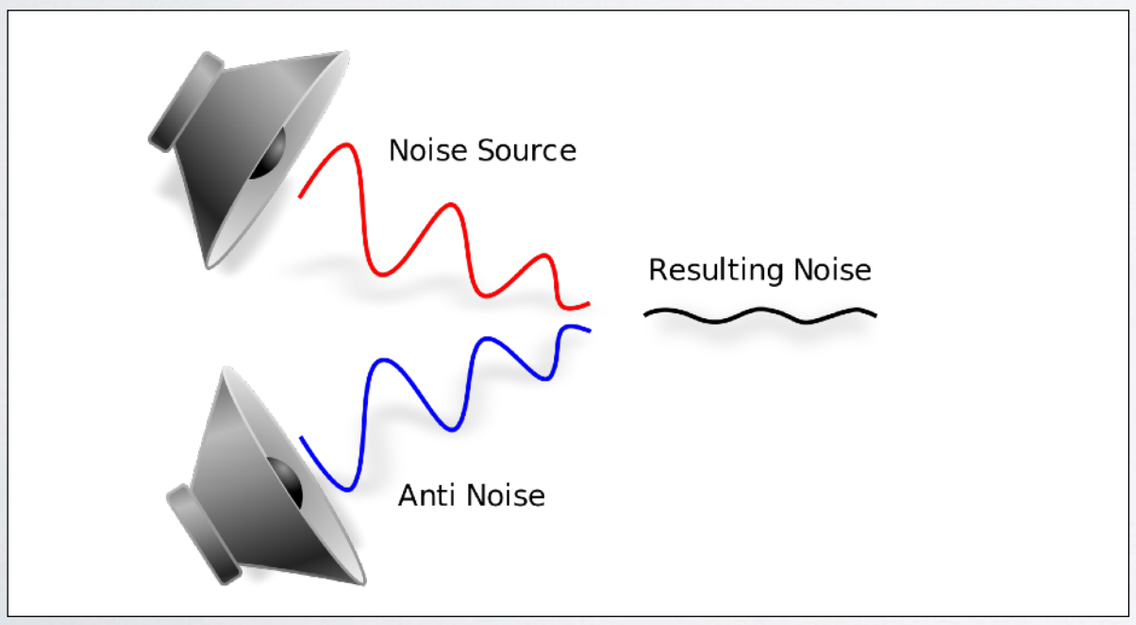
- Whenever there are wave patterns.



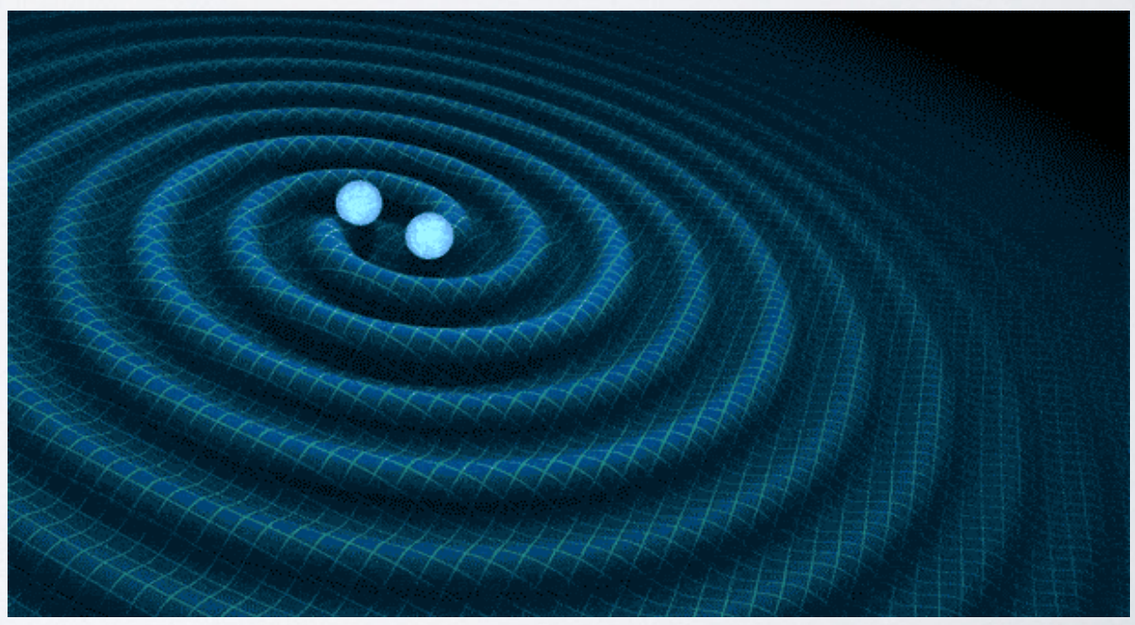
Sound equalizer, Wikipedia commons



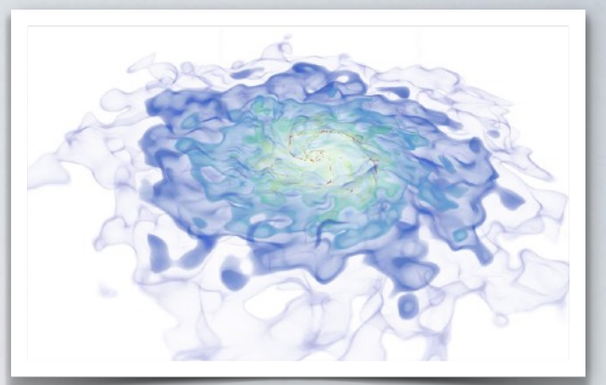
Electroencephalography (EEG), abcnews.go.com



Active noise cancelling, Wikipedia commons



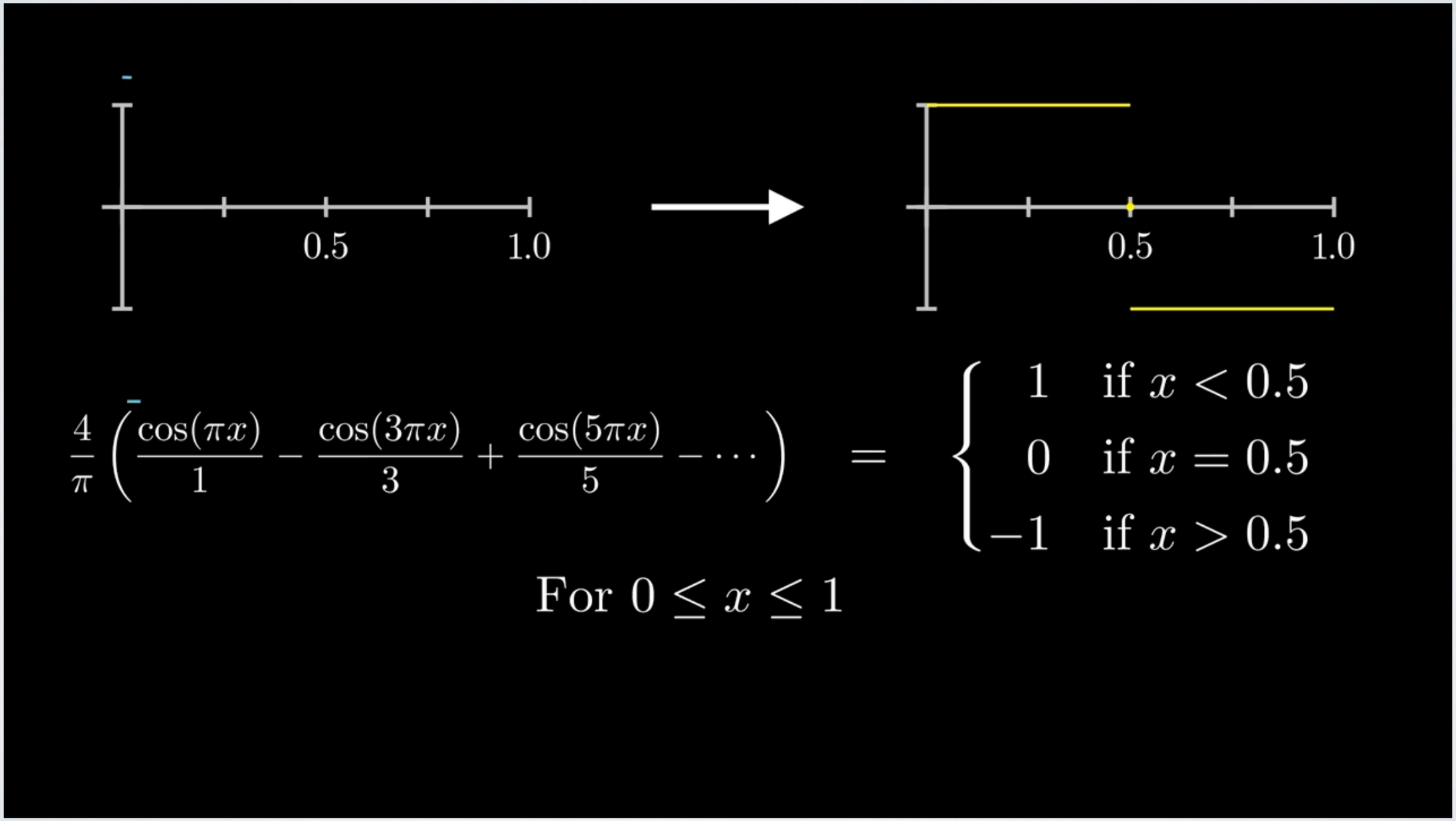
LIGO



Sine-Cosine Fourier Series

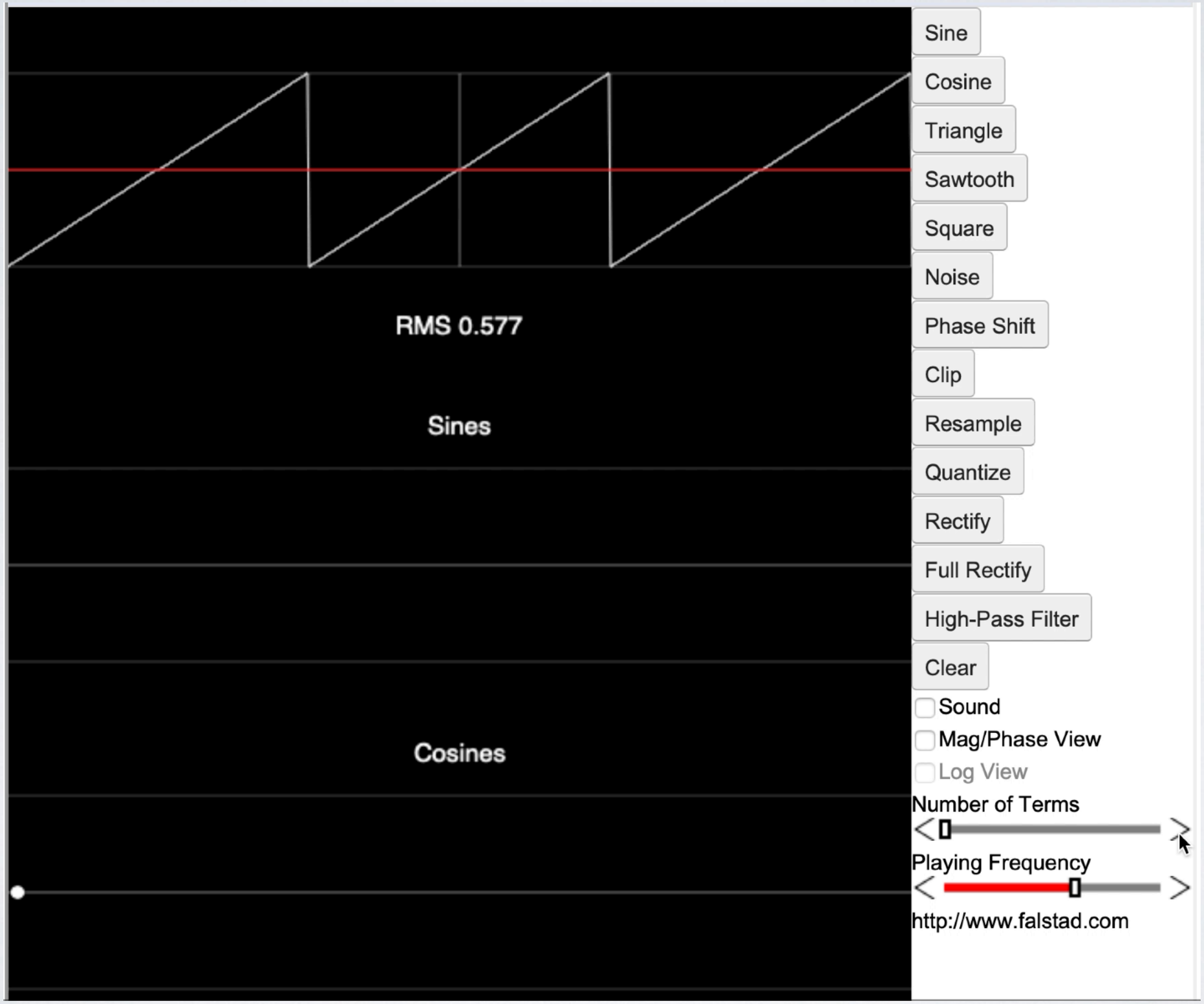
Fourier Cosine Series

- A visualization of Fourier cosine series.



youtube.com/watch?v=r6sGWTCMz2k

Fourier Sine Series



www.falstad.com/fourier/index.html

Sine-Cosine Fourier Series

- A visualization of sine-cosine Fourier series (Ch 7.5, Prob 5.7).

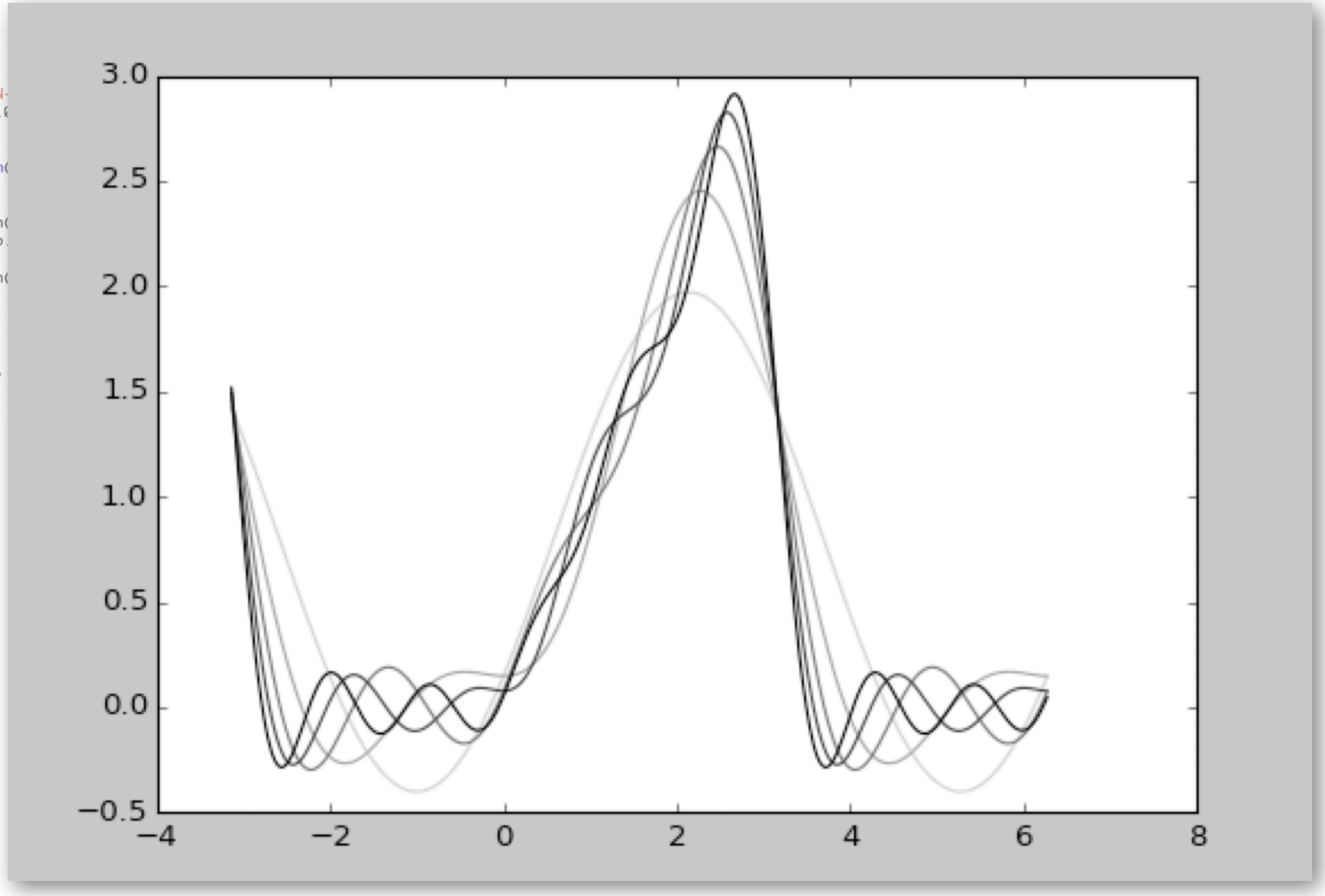
```
import numpy as np
import matplotlib.pyplot as plt

N = 50 # partial sums up to the N-
t = np.arange(-np.pi, 2*np.pi, 0.0

def my_fourier(n, t):
    partial_sum = [np.pi/4] * len(
    for nn in range(1, n+1):
        if nn%2==1:
            partial_sum += np.sin(
            partial_sum -= (2./np.
        else:
            partial_sum -= np.sin(
    return partial_sum

for n in range(N, N+1):
    col = 1. - float(n)/float(N)
    plt.plot(t, my_fourier(n, t),

plt.show()
```



Sine-Cosine Fourier Series

- A visualization of sine-cosine Fourier series (Ch 7.5, Prob 5.7).

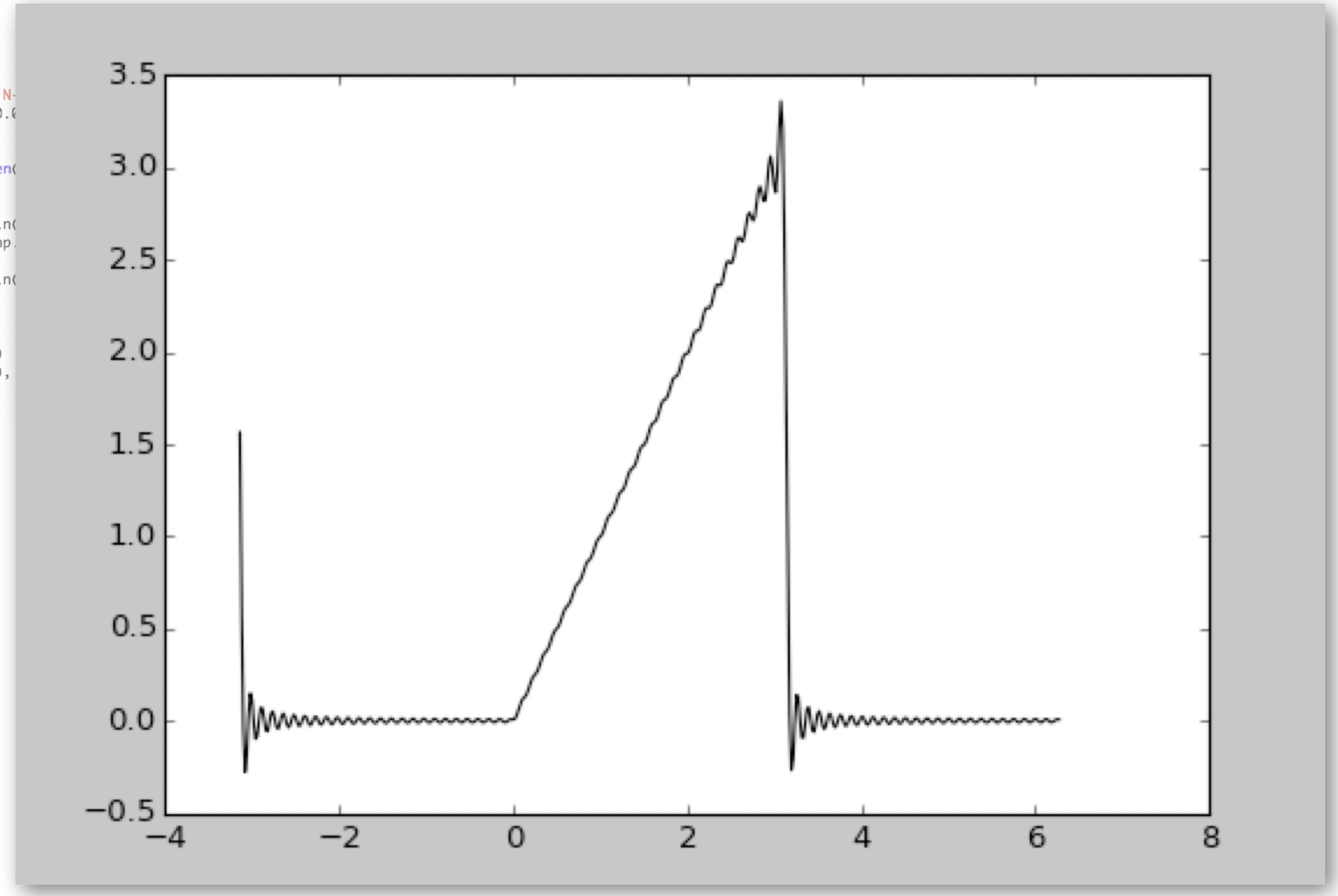
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plt.show()
```



Sine-Cosine Fourier Series

- A visualization of sine-cosine Fourier series (Ch 7.5, Prob 5.7).

WolframAlpha computational intelligence

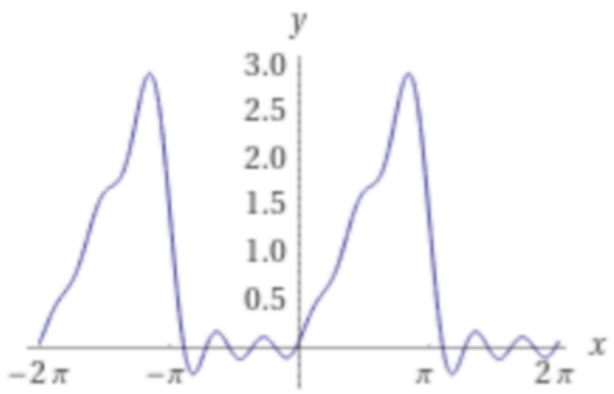
pi/4 - (2/pi) * [cos(x) + cos(3x)/3^2 + cos(5x)/5^2] + [sin(x) - sin(2x) / 2 + sin(3x)/3 - sin(4x)/4 + sin(5

NATURAL LANGUAGE | MATH INPUT | EXTENDED KEYBOARD | EXAMPLES | UPLOAD | RANDOM

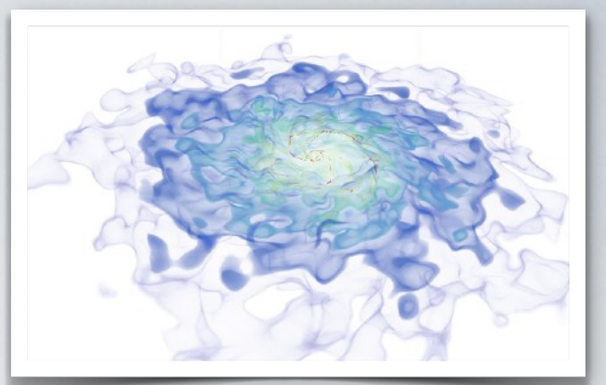
Input

$$\frac{\pi}{4} - \frac{2}{\pi}\left(\cos(x) + \frac{\cos(3x)}{3^2} + \frac{\cos(5x)}{5^2}\right) + \left(\sin(x) - \frac{1}{2}\sin(2x) + \frac{1}{3}\sin(3x) - \frac{1}{4}\sin(4x) + \frac{1}{5}\sin(5x)\right)$$

Plots



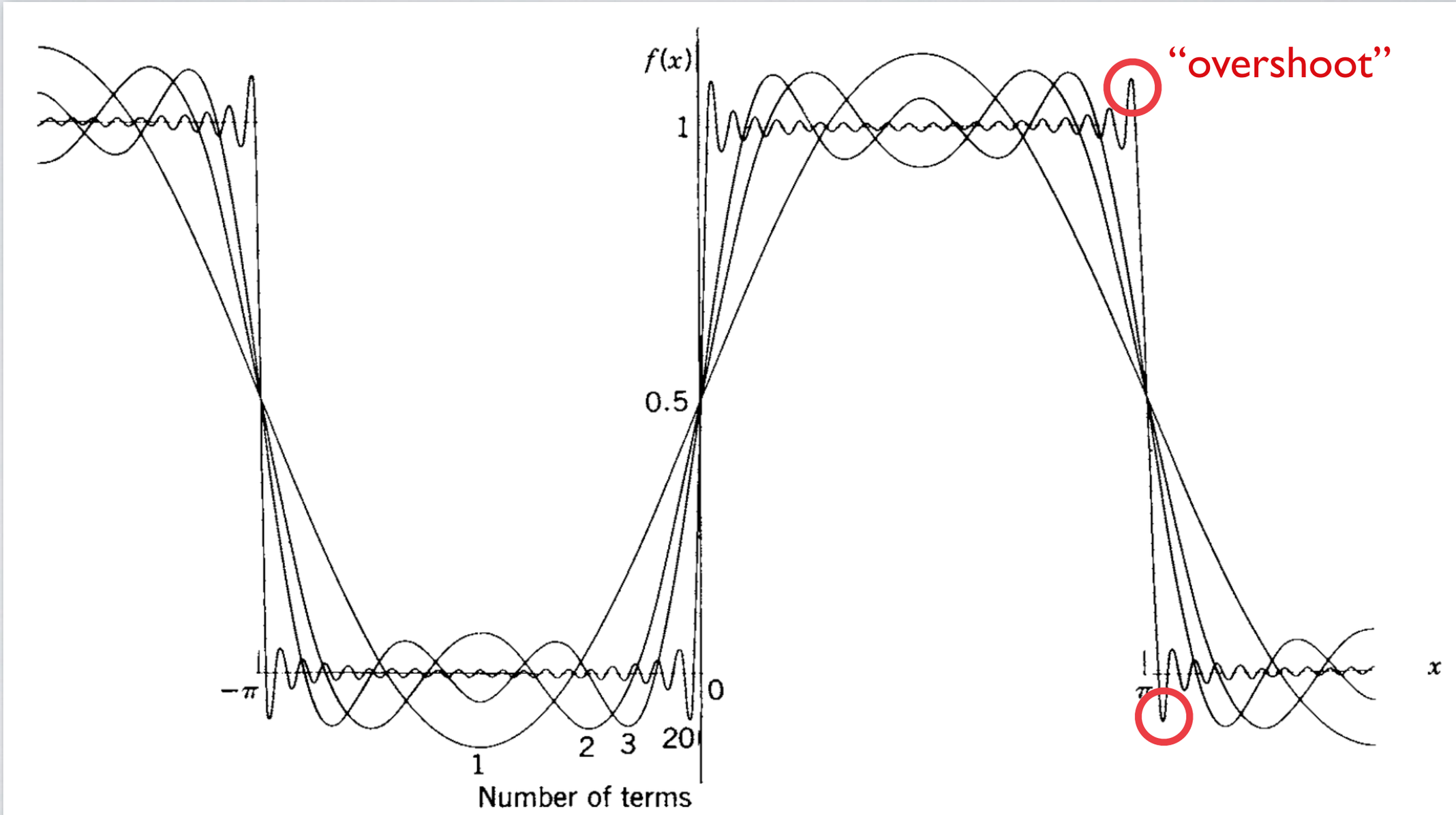
(x from −6.3 to 6.3)



Gibbs Phenomenon

Fourier Sine Series

- A visualization of Fourier sine series (Ch 7.6, Figure 6.2).



Sine-Cosine Fourier Series

- A visualization of sine-cosine Fourier series (Ch 7.5, Prob 5.7).

```
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plt.show()
```

